Department of Computer Science and Engineering
Academic Year 2021– 2022(Odd Semester)

Degree, Semester & Branch: B.E, V & CSE
Course Code & Title: CS8501 & Theory of Computation
Name of the Faculty member (s): Dr. I Gethzi Ahila Poornima

Innovative Practice Description

- **Unit / Topic: Unit II / Minimization of Finite Automata**
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO 3
- **Activity Chosen:** Active Learning- Learning by Doing
- **Justification:**

  Minimization of Finite Automata is essential because size of automata will cause more cost. Also, the way to minimize the number of states is handled in different way by different authors. Hence this topic is very crucial where many will commit mistakes in solving this problem.
- **Time Allotted for the Activity:** 30 Minutes
- **About the Event:**

  A particular topic is chosen and will be intimated to all the students either through email or in LMS. Based on the topic chosen, the announcement will be made earlier.
- **Details of the Implementation:**

  The students were intimated about this event and topic one week in prior so that they could find enough time to prepare for the topic. Randomly a student was called to present his/her topic in board or using power point presentation. For this event Ms. BhujaSri of this class solved the problem of minimization of the given DFA using Table filling algorithm.

  First, she explained about the simplification of DFA by eliminating unreachable and dead states. Then the construction of equivalence table for the given DFA was explained in white board. A few sample pictures taken during the event (solved by Ms.BhujaSri) are shown in Figure 1.
- **CO – PO / PSO mapping:**

CO	PO2	PO3	PSO1
CO5	1	2	2

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO2	PO3	PSO1
	1	2	2

Justification for correlation	Students could formulate complex engineering problems by expressing them in a standard notation such as regular expression and automata which models the system	Students will be able to minimize the finite automata which is useful for compiler.	By formulating regular expressions and FA, students will be able to develop various software components especially system software components

- **Images / Screenshot of the practice:**


Figure 1: Picture taken during the event

- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - Most of the students felt that this event helped them to get practice in solving the minimization problem.
  - ❖ ***Benefit of the practice:***
    - They can easily remember this concept by teaching and solving in the white board.
    - Also, they were well prepared for the queries that might be asked in the session.
  - ❖ ***Challenges faced in implementation:***
    - Few students faced difficulties in filling the non-equivalence cells.

Signature of Faculty Member **HOD**